

Chapter 4, Data preparation for the 2017-05-30 data

The purpose to this notebook, is to import the 2017-05-30 data and transform it so that it can be analyzed more easily.

While the 2016 data sets were filtered at a threshold of either 65 or 125, this data set does not have a reported threshold value. Consequently, the data is analyzed on it's own.

Imports and Initialization

Code Imports

```
In [1]: import pandas as pd
import numpy as np
import collections
```

Import the raw data

Note: For speed of loading, the original xlsx files were already transformed to csv

```
In [2]: df3 = None
df3 = pd.read_csv('2017-05-30.csv')
```

Extract the 'Totals' columns to a new dataframe

```
In [3]: totalsXX = None
totalsXX = pd.DataFrame(df3.loc[df3[' Number of Cluster (Total)'].notnull()])
```

Drop unused columns, fix column names, fix column types, and clean up the index

```
In [4]: #Drop
for col_name in ['ROI Index ', 'Individual Cluster Area (pixels)', 'Individual Cluster Intensity',
                'Circularity of Cluster']:
    if col_name in totalsXX.columns:
        del totalsXX[col_name]

#Fix names
totalsXX.rename(columns = {'ImageName ': 'Image Name',
                          ' Time Point ': 'Time Point',
                          ' Number of Cluster (Total)': 'Total Number of Clusters',
                          ' Area of Protein Cluster (Total in pixels)': 'Total Area of Protein Clusters',
                          ' Intensity of Protein Cluster (Total)': 'Total Intensity of Protein Clusters' },
                inplace=True)

#Fix types
totalsXX['Total Number of Clusters'] = totalsXX['Total Number of Clusters'].apply(np.int)
totalsXX['Total Area of Protein Clusters'] = totalsXX['Total Area of Protein Clusters'].apply(np.int)
totalsXX['Total Intensity of Protein Clusters'] = totalsXX['Total Intensity of Protein Clusters'].apply(np.int)

#Fix index
totalsXX = totalsXX.reset_index(drop=True)
```

Create helper columns with scaled scores

```
In [5]: max_value = totalsXX['Total Number of Clusters'].max()
totalsXX.insert(loc=5, column='Total Number Scaled',
               value=round(((totalsXX['Total Number of Clusters'] / max_value) * 100), 2))

max_value = totalsXX['Total Area of Protein Clusters'].max()
totalsXX.insert(loc=6, column='Total Area Scaled',
               value=round(((totalsXX['Total Area of Protein Clusters'] / max_value) * 100), 2))

max_value = totalsXX['Total Intensity of Protein Clusters'].max()
totalsXX.insert(loc=7, column='Total Intensity Scaled',
               value=round(((totalsXX['Total Intensity of Protein Clusters'] / max_value) * 100), 2))
```

Create a helper column for receptor names

```
In [6]: image_list = set(totalsXX['Image Name'])

all_prefixes = [i.split('-',2) for i in image_list]
prefixes = set([p[0] for p in all_prefixes])

for p in prefixes:
    totalsXX.loc[totalsXX['Image Name'].str.contains(p), 'Receptor'] = p
```

Create a helper column to group results by experiment step

```
In [7]: # Note: B1M29 reverses the order of the -l/-nl pre/post labeling.
steps = [('-l-pre', '-l-pre'), ('-l-post', '-l-post'), ('-nl-pre', '-nl-pre'),
         ('-nl-post', '-nl-post'),
         ('-pre-l', '-l-pre'), ('-post-l', '-l-post'), ('-pre-nl', '-nl-pre'),
         ('-post-nl', '-nl-post')]

for s in steps:
    totalsXX.loc[totalsXX['Image Name'].str.contains(s[0]), 'Experiment Step'] = s[1]
```

Create a helper column to group results by experiment run

```
In [8]: run = [('-1_', '1'), ('-2_', '2')]

for r in run:
    totalsXX.loc[totalsXX['Image Name'].str.contains(r[0]), 'Run Group'] = r[1]
```

Fix data types

```
In [9]: totalsXX['Total Number Scaled'] = totalsXX['Total Number Scaled'].apply(np.int)
totalsXX['Total Area Scaled'] = totalsXX['Total Area Scaled'].apply(np.int)
totalsXX['Total Intensity Scaled'] = totalsXX['Total Intensity Scaled'].apply(np.int)
totalsXX['Run Group'] = totalsXX['Run Group'].apply(np.int)
```

Output dataframe to CSV

```
In [10]: totalsXX.to_csv('TXX_Totals.csv', index=False)
```